

Approval Workflow Engine - Backend Engineering Case Study

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Problem Statement

Many approval processes are still handled manually through email, spreadsheets, chat messages, or disconnected tools. This creates delays, inconsistent decision-making, poor visibility into request status, and weak audit trails. As organizations grow, these manual workflows become harder to manage and more error-prone.

Organizations need approval workflows for expense reports, purchase orders, time-off requests, and other business processes. Your task is to build a generic, reusable approval workflow engine.

The problem is to build a generic, reusable approval workflow engine that can support multiple accounting and business use cases without hardcoding each approval process. The system should make it easy to define workflows, submit approval requests, route them to the correct approvers, capture approve/reject decisions, and preserve a reliable history of who made each decision and when.

Overview

Design and implement an agentic AI-powered assistant and backend service for a flexible approval workflow engine that can be integrated into various business applications. This engine will allow organizations to define custom approval processes, route requests through multiple approvers, and track decision history.

Core Capabilities Needed

1. Dynamic Workflow Definition – Create multi-step approval workflows with conditional routing
2. Request Submission – Submit items for approval with metadata
3. Decision Making – Approve, reject, or request changes with comments
4. Delegation – Temporary delegation of approval authority
5. Audit Trail – Complete history of all actions taken
6. Agentic Interaction – Users can interact with an AI assistant to complete workflow tasks

Timeline & Expectations

Day Before(Friday):

- Review this case study scenario
- Identify questions you have about data models, constraints, edge cases, etc.
- Prepare your development environment with your preferred stack

Case Study Day(Weekend):

- Morning Q&A session with the team
- Build the API backend independently using your preferred stack
- Use AI-assisted tools for rapid development
- Write unit tests for core functionality

Next Day(Monday):

- Submit final code and documentation
- Demo and architecture review session

Part 1: Data Model Design

Design a database schema that supports:

- Workflow Templates
- Workflow Instances
- Approval Steps
- Conditions
- Approvers (users and groups)
- Delegations
- Decisions
- Audit Trail

Deliverables:

- SQL schema with relationships and indexes
- Explanation of design decisions
- Consider dynamic routing, parallel approvals, group approvals, and escalation rules

Part 2: User Stories & API Implementation

Implement RESTful APIs for:

- Workflow administration
- Request submission
- Approval inbox/dashboard
- Decision making
- Delegation management

Requirements:

- Authentication and authorization
- Validation and error handling
- Pagination for list operations
- API documentation

Part 3: Agentic Workflow Assistant

Build an AI-powered assistant that enables users to interact with the approval workflow system through natural language.

Example Scenarios

1. Submit a Request

User: 'I need approval for a \$5,000 laptop purchase.'

Assistant:

- Collects missing information
- Selects the appropriate workflow
- Submits the request
- Returns status and next approvers

2. Review Pending Approvals

User: 'What requests are waiting for my approval?'

Assistant:

- Retrieves pending approvals
- Includes delegated approvals
- Summarizes key information

3. Make an Approval Decision

User: 'Approve Sarah's laptop purchase request.'

Assistant:

- Verifies authorization
- Presents context
- Requests confirmation
- Executes the approval

4. Create a Workflow Template

User: 'Create a workflow where purchases over \$10,000 require Finance approval followed by VP approval.'

Assistant:

- Generates a workflow definition
- Validates the configuration
- Presents the workflow for review
- Creates it after confirmation

Technical Requirements

Agent Endpoint

Example:

POST /agent/chat

Tool Calling

The assistant should interact with the workflow engine through service interfaces or tool calls rather than directly modifying data.

Example tools:

- Create workflow template
- Submit request
- Retrieve pending approvals
- Approve request
- Reject request
- Create delegation
- Retrieve audit history

Multi-turn Conversations

Support conversations that require gathering missing information.

Authorization & Safety

- Respect role-based permissions
- Prevent unauthorized actions
- Require confirmation before irreversible actions
- Record all actions in the audit trail

Observability

Provide visibility into:

- User requests
- Agent reasoning or orchestration steps
- Tool invocations

- Failures and retries

Deliverables

- Agent API implementation
- Tool definitions or service interfaces
- Example conversations
- Tests covering at least one multi-turn workflow
- Documentation describing architecture, tool-calling strategy, authorization, and failure handling

Evaluation Criteria

1. API Design
2. Code Quality
3. Testing
4. Reliability
5. Agent Design
6. Tool Usage
7. User Experience
8. Security
9. Observability
10. AI Leverage

Submission Requirements

Upload to GitHub with the following:

Repository Structure:

- Source code
- Database schema and migrations
- Test suite

Documentation:

README.md

SOLUTION.md

Include:

- setup instructions,
- architecture decisions,
- AI usage,
- tradeoffs,
- production-readiness considerations,
- and lessons learned.